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Database Security:

Authentication, Authorization and access control, DAC, MAC and RBAC models, Intrusion detection, SQL injection.

Advanced topics: Object oriented and object relational databases, Logical databases, Web databases, Distributed databases, Data warehousing and data mining

Database Security

Authentication

- Authentication is the process of identifying someone's identity by assuring that the person is the same as what they are claiming for.
- It is used by both server and client.
- The server uses authentication when someone wants to access the information
- The server needs to know who is accessing the information
- The authentication by the server is done mostly by using the username and password.

Authentication Factors

- Single-factor authentication is the simplest way of authentication.
- It needs a username and password to allow a user to access a system
- Two-factor Authentication needs two-step verification to authenticate a user eg: verify the user by sending the OTP
- Multi-factor Authentication is the most secure and advanced level of authorization.
- Multi-factor Authentication requires two or more than two levels of security from different and independent categories eg: financial organizations, banks

Authentication techniques

1) Password-based authentication

- It is the simplest way of authentication.
- It requires the password for the particular username.
- If the password matches with the username and both details match the system's database, the user will be successfully authenticated.

2) Passwordless authentication

- In this technique, the user doesn't need any password
- An OTP (One-time password) or link on registered mobile number or phone number. It can also be said OTP-based authentication.

3) **Social Authentication**

- Social authentication does not require additional security
- it verifies the user with the existing credentials for the available social network.

Authorization

- Authorization is the process of granting someone to do something.
- It means it a way to check if the user has permission to use a resource or not.
- It defines that what data and information one user can access. It is also said as AuthZ.
- The authorization usually works with authentication so that the system could know who is accessing the information.

Authorization Techniques

1) Role-based access control

RBAC or Role-based access control technique is given to users as per their role or profile in the organization.

- It can be implemented for system-system or user-to-system.

2) JSON web token

- JSON web token or JWT is an open standard used to securely transmit the data between the parties in the form of the JSON object.
- The users are verified and authorized using the private/public key pair.

3) **SAML**

- SAML stands for **Security Assertion Markup Language**.
- It is an open standard that provides authorization credentials to service providers.
- These credentials are exchanged through digitally signed XML documents.

4) **OpenID authorization**

- It helps the clients to verify the identity of end-users on the basis of authentication

Authentication	Authorization
Authentication is the process of identifying a user to provide access to a system.	Authorization is the process of giving permission to access the resources.
the user or client and server are verified.	It is verified that if the user is allowed through the defined policies and rules.
It is usually performed before the authorization.	It is usually done once the user is successfully authenticated.
It requires the login details of the user, such as user name & password, etc	It requires the user's privilege or security level.
Eg: Entering Login details is necessary for the employees to authenticate themselves to access the organizational emails or software.	Eg: After employees successfully authenticate themselves, they can access and work on certain functions only as per their roles and profiles.

Intrusion detection

- An Intrusion Detection System (IDS) is a security tool that monitors a computer network or systems for malicious activities or policy violations
- It helps detect unauthorized access
- Maintaining network security and protecting sensitive data from cyber-attacks.
- The main duties of an Intrusion Detection System (IDS) are anomaly detection and reporting
- An IDS (Intrusion Detection System) monitors the traffic on a computer network to detect any suspicious activity.

- The system administrator can then investigate the alert and take action to prevent any damage or further intrusion.
- IDS can detect any suspicious activities and alert the system administrator before any significant damage is done.
- IDS identifies potential threats early, allowing for quicker response to prevent damage.

SQL injection

- SQL injection is a code injection technique that might destroy your database.
- SQL injection is one of the most common web hacking techniques.

Object oriented and object relational databases

1) Object-oriented database system

- Extension of an object-oriented programming language
- Includes failure recovery, transaction management and query processing
- Features such as inheritance make it possible to use existing code thereby shortening development time.
- The efficiency of query processing is low.

2) **object-relational database system**

- Extended to incorporate object-oriented characteristics.
- An object-relational database is one that is based on both the relational and object-oriented database models.
- Processing of queries is quite effective.

Logical Database

- A Logical Database is a special type of ABAP (Advance Business Application and Programming)
- Used to retrieve data from various tables and the data is interrelated to each other.
- logical database provides a read-only view of Data.
- A Logical database uses only a hierarchical structure of tables
- Data is organized in a Tree-like Structure and the data is stored as records that are connected to each other through edges (Links).



RETRIEVING DATA



STUDENT DETAILS

Web Databases

- A web database is a system for storing and displaying information that is accessible from the Internet / web.
- The database might be used for any of a wide range of functions, such as a membership database, client list or inventory database.
- A web database is a type of Web Application.
- A web database is ideal for situations when the information should be shared, or when it must be accessed from various locations

Distributed databases

- A distributed database is a database that is not limited to one system, it is spread over different sites
- Multiple computers or over a network of computers
- In a homogeneous database, all different sites store database identically.
- In a heterogeneous distributed database, different sites can use different schema and software that can lead to problems in query processing and transactions.
- It is used in Corporate Management Information System.
- It is used in multimedia applications